Mathematical Proof: Pythagorean Theorem

Statement

In a right triangle, the square of the hypotenuse equals the sum of squares of the other two sides: $a^2 + b^2 = c^2$

Proof

Consider a right triangle with sides a, b, and hypotenuse c.

Construct a square with side length $(a + b)$ and arrange four copies of the triangle inside it.

The area of the large square is $(a + b)^2$, which equals the area of four triangles plus the area of the inner square: $4(ab/2) + c^2$

Simplifying: $a^2 + 2ab + b^2 = 2ab + c^2$

Therefore: $a^2 + b^2 = c^2$ ✓